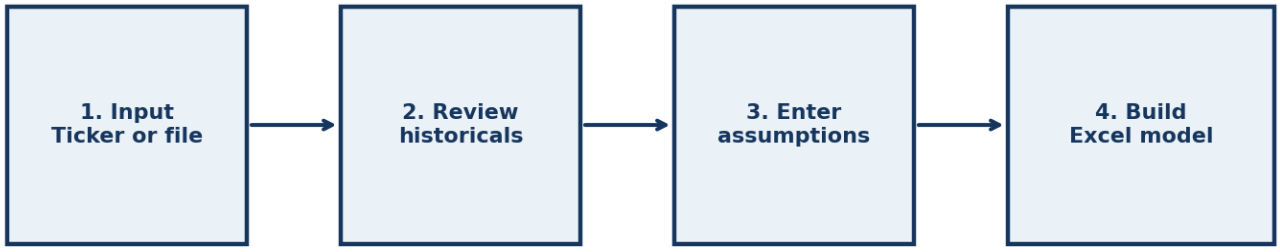
Generic DCF Valuation Automation

Technical Documentation • Template-independent Python and Excel model

Purpose

This project builds a DCF valuation workbook programmatically, without relying on an existing workbook template. It supports annual historical financials from yfinance or an uploaded CSV/XLSX file, prompts for analyst assumptions, and creates a formatted five-sheet Excel model.



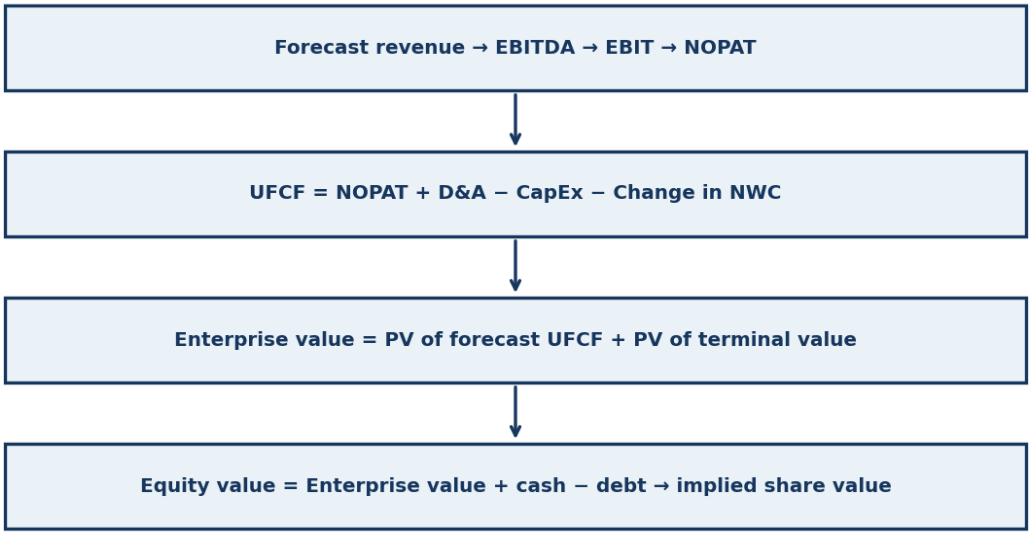
1. Architecture

Input layer: ticker or historical file. **Mapping layer:** maps common financial statement labels into standardized line items. **Assumption layer:** collects forecast and valuation inputs. **Model layer:** writes linked formulas into the workbook. **Presentation layer:** applies styles, charts, sensitivity formatting and dashboard KPI cards.

2. Workbook structure

Sheet	Purpose
Financials	Historical annual financials, source label and notes for missing values.
Assumptions	Editable assumptions, cost of equity, after-tax cost of debt and WACC.
Forecast	Forecast income statement and unlevered free cash flow.
DCF & Sensitivity	DCF valuation, implied share value, upside/downside and WACC/terminal-growth sensitivity.
Charts & Dashboard	Valuation KPI cards, revenue vs EBITDA chart, UFCF vs PV of UFCF chart.

3. Core formulas



Cost of equity: Risk-free rate + beta × equity risk premium.

After-tax cost of debt: Pre-tax cost of debt × (1 – tax rate).

WACC: Cost of equity × equity weight + after-tax cost of debt × debt weight.

EBIT: EBITDA – D&A. **NOPAT:** EBIT – cash taxes on EBIT.

UFCF: NOPAT + D&A – CapEx – change in net working capital.

Terminal value: Final forecast UFCF × (1 + terminal growth) / (WACC – terminal growth).

Enterprise value: PV of forecast UFCF + PV of terminal value.

Equity value: Enterprise value + cash – debt. **Implied share value:** Equity value / shares outstanding.

4. Sensitivity table

The sensitivity grid varies WACC and terminal growth around a base case and calculates implied value per share. The terminal growth rate must remain below WACC; otherwise the perpetuity formula is invalid. Review the grid's ranges before interpreting results.

5. Data limitations and model controls

yfinance labels and coverage differ across companies. Some data may be unavailable or mapped imperfectly; some providers return cash flow or balance sheet figures in different units from price/share data. The analyst must confirm units, fiscal-year dates, shares outstanding, cash, debt, tax, CapEx and working capital. The model does not guarantee that every input is available or correct.

The current implementation is a general starter model. In particular, the uploaded file should have one row per fiscal year and columns with common names such as Revenue, EBITDA, Total Debt, Cash & Cash Equivalents and Shares Outstanding. Use the included sample CSV as a format reference.

6. Dependencies and validation

Python 3.10 or later is recommended. Install packages with `pip install -r requirements_generic_dcf.txt`. Ticker mode requires internet access and yfinance. Uploaded-file mode does not need yfinance for retrieval, although the package is listed in requirements for both modes.

The script has been syntax-compiled and the generated sample workbook has been checked for the five expected sheets, formula cells and two charts. Excel formula values are recalculated by Excel or LibreOffice when opened; openpyxl writes

formulas but does not calculate them itself.

7. Known improvements for production use

For professional transaction work, consider adding more robust company-specific line-item mapping, quarterly-to-annual normalization, explicit units and currency conversion, separate diluted share count and net debt reconciliation, a debt schedule, a full sensitivity formula audit, model error flags, and tests against a known valuation case. Review every model output before use.